Quiz 1: Instructor Key

September 15, 2026

Staff only. Five parts, 2 points each, 10 points total. Award 2 points for a correct final answer and 0 otherwise.

1(a)	1(b)	1(c)	2(a)	2(b)
3	9	7	D	A

1. State and action values

6 points

In (a)–(b), π chooses `serve`; $\gamma = \frac{2}{3}$ in all three parts.

(a) The cold cache

Serving in C pays 1 and returns to C with probability 1:

$$V^\pi(\mathrm{C}) = 1 + \tfrac{2}{3}V^\pi(\mathrm{C}) \quad \Longrightarrow \quad \boxed{V^\pi(\mathrm{C}) = 3}.$$

C continues to earn rewards under this policy; its value is not zero.

(b) The hot cache

The Bellman equation averages over both successors:

$$\begin{aligned} V^\pi(\mathrm{H}) &= 5 + \tfrac{2}{3}\left(\tfrac{1}{2}V^\pi(\mathrm{H}) + \tfrac{1}{2}V^\pi(\mathrm{C})\right) \\ &= 5 + \tfrac{1}{3}V^\pi(\mathrm{H}) + 1. \end{aligned}$$

Thus $\frac{2}{3}V^\pi(\mathrm{H}) = 6$, giving $\boxed{V^\pi(\mathrm{H}) = 9}$.

(c) Independent MDP: take a once, then follow π

Action a pays -1 and leads to Y. From Y onward, the expected return under π is the supplied value 12:

$$Q^\pi(\mathrm{X}, a) = -1 + \tfrac{2}{3}V^\pi(\mathrm{Y}) = -1 + \tfrac{2}{3} \cdot 12 = \boxed{7}.$$

The immediate reward is counted once and is not discounted. The value of Y already includes all future rewards under π. No answer from (a) or (b) is used.

2. Concepts

4 points

(a) D: Planning with a known model

The model and objective are supplied, so planning is the most specific formulation. Actions have future consequences, so it is not a contextual bandit; no supervised labels are supplied. Option C describes learning through interaction with an unknown environment. Here, the model is known and no data collection is required. Random transitions do not change this classification.

(b) A: Markov state; deterministic policy allowed

The stipulated dependence of the next-state and reward distribution on the current state and action is exactly the Markov property. Random slips are compatible with it. A policy may choose a single action per state even when that action has stochastic outcomes; the full route is unnecessary here.